

Introduction

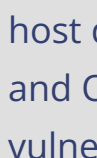
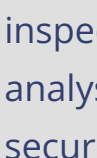
A cybersecurity lab is an isolated, controlled computing environment designed for practicing ethical hacking, penetration testing, and security research without risk to production systems or real networks. As cyber threats grow in sophistication, hands-on experience in a safe environment becomes essential for developing practical skills in vulnerability assessment, network analysis, and exploit mitigation. This report documents the complete setup of such a laboratory using Kali Linux as the primary penetration testing distribution and Oracle VirtualBox as the virtualization platform.

Ethical hacking requires a dedicated environment where tools can be deployed, network traffic can be captured and analyzed, and vulnerable applications can be safely tested. The purpose of this lab is to provide a reproducible, sandboxed workspace that mirrors real-world network conditions while maintaining full isolation from external infrastructure. This ensures that all testing activities remain legal, controlled, and educational in nature.

Objectives

O1 Install and Configure VirtualBox Set up Oracle VirtualBox as the hypervisor platform for running virtual machines with proper network adapter configuration.	O2 Deploy Kali Linux VM Import and configure Kali Linux as the primary penetration testing operating system within the virtualized environment.
O3 Configure Dual Network Adapters Set up NAT (Adapter 1) for internet access and Host-Only Adapter (Adapter 2) for isolated internal communication.	O4 Validate Connectivity and Tools Test network connectivity using ping, perform port scanning with Nmap, capture packets with Wireshark, and deploy OWASP Juice Shop via Docker.

Software and Tools Used

 Oracle VirtualBox Open-source hypervisor for creating and managing virtual machines. Enables isolated OS environments on a single physical host with configurable networking, snapshots, and resource allocation.	 Kali Linux Debian-based Linux distribution purpose-built for penetration testing and digital forensics. Ships with over 600 pre-installed tools including Nmap, Wireshark, Metasploit, and Burp Suite.
 Nmap Network exploration and security auditing tool. Performs host discovery, port scanning, service version detection, and OS fingerprinting to map network topology and identify vulnerabilities.	 Wireshark Industry-leading network protocol analyzer. Captures and inspects network packets in real-time, enabling deep analysis of traffic patterns, protocol behavior, and potential security anomalies.
 Burp Suite Web application security testing platform. Functions as an intercepting proxy between the browser and target server, allowing manual and automated testing of HTTP/HTTPS traffic for vulnerabilities.	 Docker & OWASP Juice Shop Docker provides lightweight containerization for running applications in isolated environments. OWASP Juice Shop is a deliberately vulnerable web application used for practicing web security testing and CTF challenges.

VirtualBox Installation and Configuration

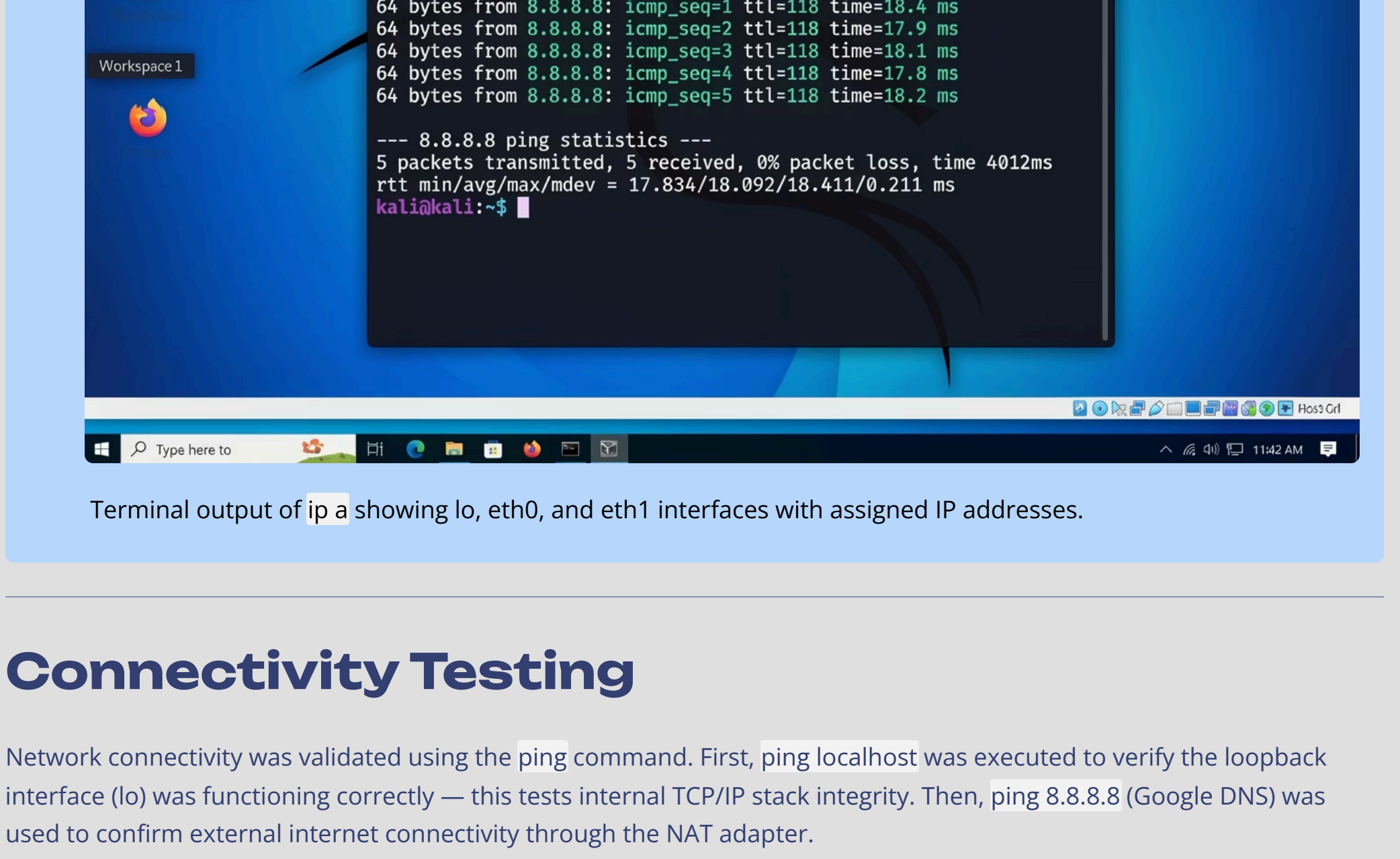
Oracle VirtualBox was installed on the host machine to serve as the virtualization platform. After installation, a new virtual machine was created with the following specifications: Type — Linux; Version — Debian (64-bit); RAM — 4GB; Hard Disk — 40GB dynamically allocated VDI. Kali Linux ISO was then attached and the VM was booted to begin the installation process.

Critical to the lab's functionality was the network adapter configuration. Adapter 1 was set to **NAT (Network Address Translation)** mode, allowing the Kali VM to access the internet through the host machine's connection. Adapter 2 was configured as a **Host-Only Adapter**, creating a private network segment between the host and the VM for isolated internal communication. This dual-adapter setup ensures both internet connectivity for tool updates and a secure, isolated testing network.

VirtualBox VM settings showing Adapter 1 (NAT) and Adapter 2 (Host-Only) configuration.

Kali Linux Setup and IP Configuration

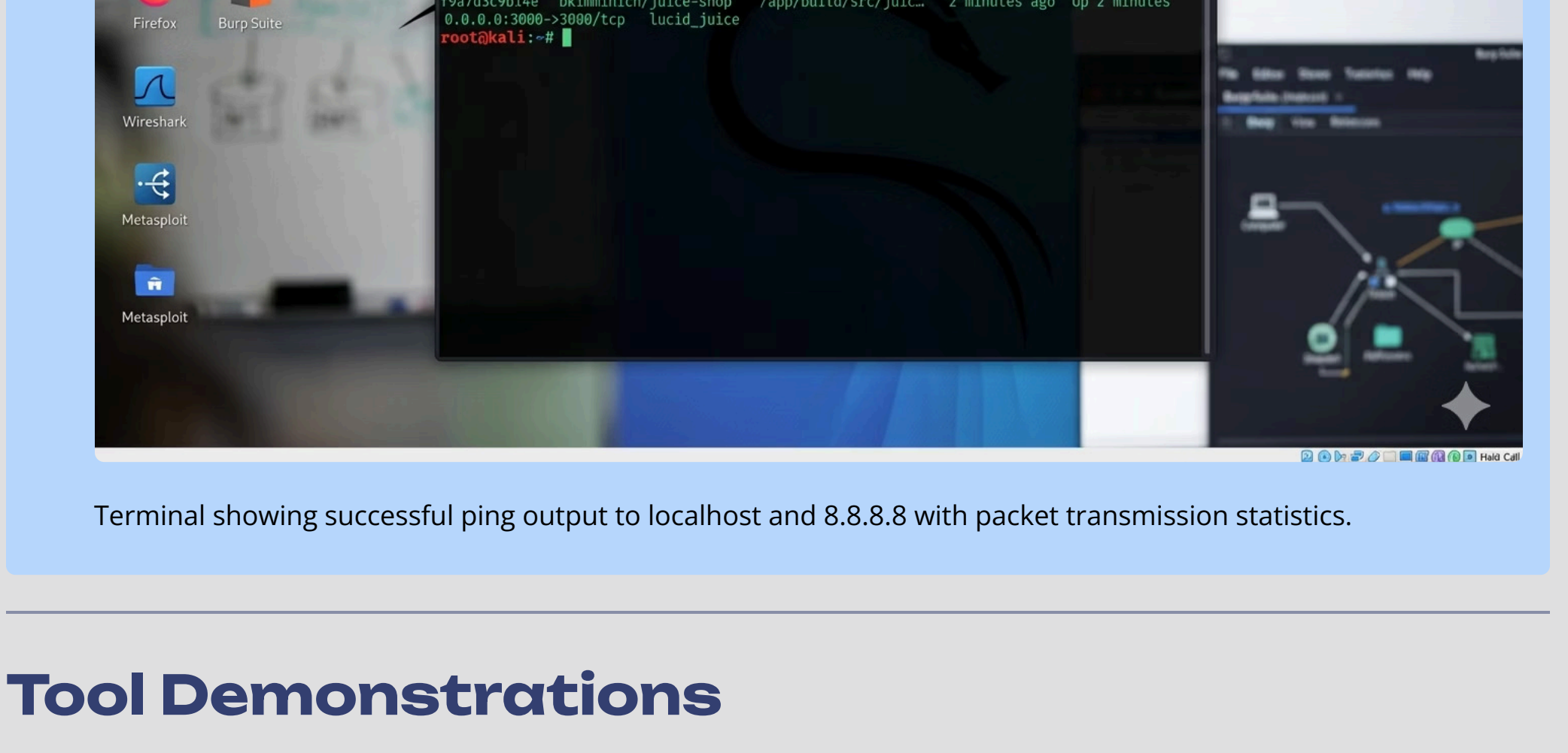
Starting the VM and Terminal Access After successful installation, the Kali Linux VM was started from VirtualBox. Upon boot, the terminal was accessed and the root terminal was enabled using <code>sudo su</code> for administrative commands. The <code>ip a</code> command was executed to display all network interfaces and their assigned IP addresses, confirming proper adapter configuration.	Key Commands Executed <pre>sudo su ip a ping localhost ping 8.8.8.8 nmap localhost</pre> Expected Output The <code>ip a</code> command should show <code>eth0</code> with a NAT IP (typically 10.0.2.x) and <code>eth1</code> with a Host-Only IP (typically 192.168.56.x). Successful ping to 8.8.8.8 confirms internet connectivity through the NAT adapter.
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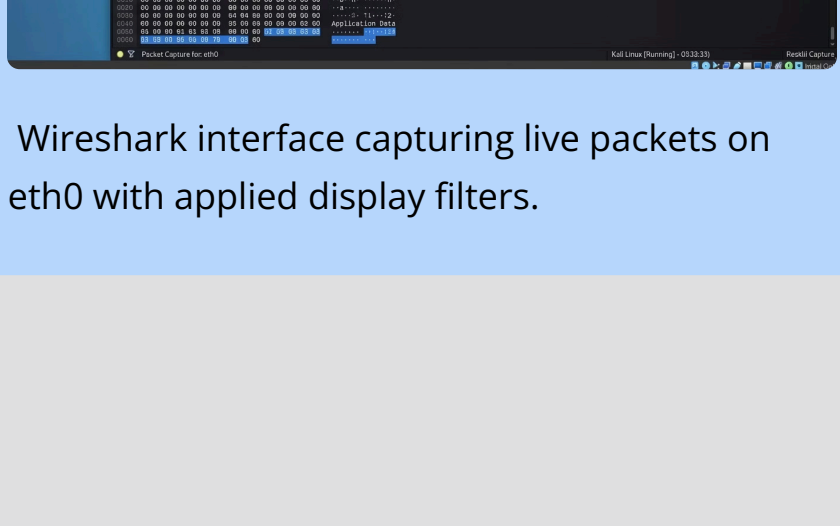
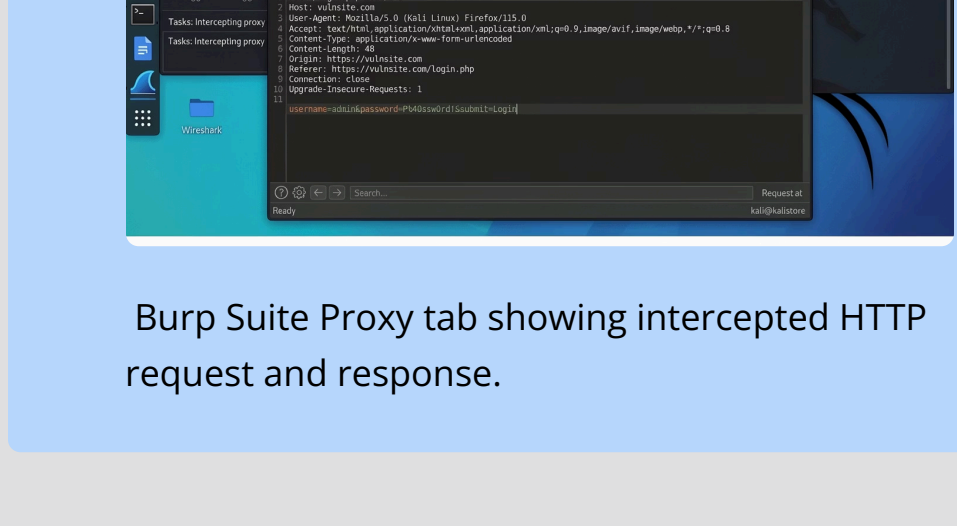
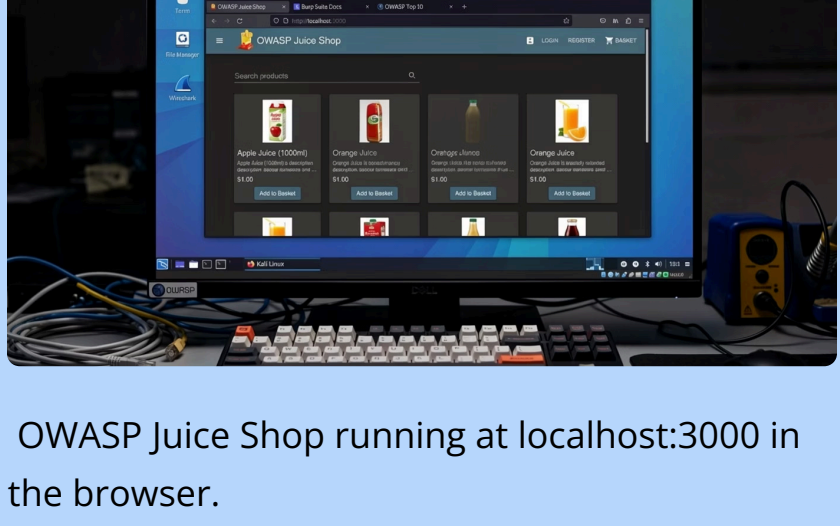
Connectivity Testing

Network connectivity was validated using the `ping` command. First, `ping localhost` was executed to verify the loopback interface (`lo`) was functioning correctly — this tests internal TCP/IP stack integrity. Then, `ping 8.8.8.8` (Google DNS) was used to confirm external internet connectivity through the NAT adapter.

Successful packet transmission with low latency and zero packet loss confirmed that both network adapters were correctly configured. The NAT adapter successfully routed traffic to the external network, while the Host-Only adapter maintained the isolated internal network segment. This dual-connectivity setup is foundational for ethical hacking workflows that require both internet access (for tool updates, package installation) and isolated testing (for safe exploit practice).



Tool Demonstrations

Nmap — Port Scanning & Host Discovery Nmap was used to scan the <code>localhost</code> environment to identify open ports, running services, and potential entry points. The command <code>nmap localhost</code> performed a basic TCP port scan, revealing services listening on the local machine. Nmap's output includes port numbers, service names, and state (open/closed/filtered), providing a foundational network map for further penetration testing activities. <pre>nmap localhost nmap -sV 192.168.56.1</pre>	Wireshark — Packet Capture & Analysis Wireshark was launched and configured to capture packets on the <code>eth0</code> interface. Live traffic monitoring revealed HTTP requests, DNS queries, TCP handshakes, and other protocol interactions. Filters such as <code>tcp.port == 80</code> and <code>dns</code> were applied to isolate specific traffic types. This exercise demonstrated how network analysts can inspect raw packet data to detect anomalies, analyze protocol behavior, and identify potential security issues. 
Burp Suite — Web Proxy Interception Burp Suite was configured as an intercepting proxy (default port 8080) to capture and modify HTTP/HTTPS traffic between the browser and web applications. The browser was configured to route traffic through Burp's proxy, and the CA certificate was installed to enable HTTPS interception. This setup allows ethical hackers to manually inspect, modify, and replay web requests to test for vulnerabilities such as SQL injection, XSS, and insecure direct object references. 	Docker & OWASP Juice Shop Docker was installed on Kali Linux using <code>apt install docker.io</code> and the service was started with <code>systemctl start docker</code> . The OWASP Juice Shop vulnerable application was then deployed using: <pre>docker run -d -p 3000:3000 bkimminich/juice-shop</pre> The application was accessed at <code>localhost:3000</code> , providing a realistic, intentionally vulnerable web application for practicing web security testing techniques including authentication bypass, SQL injection, and XSS attacks. 

Challenges Faced

Internet Connectivity in Kali VM Initial NAT adapter misconfiguration prevented internet access. Resolved by verifying Adapter 1 was set to NAT mode and restarting the network manager service with <code>systemctl restart NetworkManager</code> .	Network Adapter Configuration Setting up the dual-adapter configuration required careful attention to VirtualBox network settings. Adapter 2 (Host-Only) required enabling the VirtualBox Host-Only Network interface on the host machine through <code>File → Preferences → Network</code> .
DHCP IP Assignment The <code>eth1</code> Host-Only interface occasionally failed to receive a DHCP-assigned IP. This was resolved by manually assigning a static IP within the Host-Only network range (192.168.56.x) using <code>ip addr add</code> commands.	VirtualBox Extension Pack Certain features such as USB 2.0/3.0 support and improved network performance required the VirtualBox Extension Pack to be installed separately, matching the exact VirtualBox version in use.

Learning Outcomes

Virtualization Gained hands-on experience with VirtualBox hypervisor, VM creation, resource allocation, snapshot management, and isolated environment design for security testing.	Network Configuration Understood NAT, Host-Only, and Bridged networking modes, IP addressing, interface management, and how network topology affects penetration testing capabilities.
Packet Analysis Learned to capture, filter, and interpret network packets using Wireshark, gaining insight into TCP/IP protocols, HTTP traffic, and network-level security monitoring.	Penetration Testing Tools Practiced using Nmap for reconnaissance, Burp Suite for web application testing, and Docker for deploying vulnerable environments — foundational skills for ethical hacking.

Conclusion

This internship task successfully demonstrated the end-to-end setup of a professional cybersecurity laboratory environment using Kali Linux and Oracle VirtualBox. The lab provides a safe, isolated, and reproducible platform for practicing ethical hacking techniques, network analysis, and web application security testing. Through this exercise, a complete virtualization infrastructure was established, network adapters were properly configured for both internet access and internal isolation, and a suite of industry-standard security tools — including Nmap, Wireshark, Burp Suite, Docker, and OWASP Juice Shop — were installed, configured, and validated.

The hands-on experience gained through this lab setup forms a critical foundation for advanced cybersecurity studies. Understanding how to create and manage isolated testing environments is a prerequisite skill for any aspiring penetration tester or security analyst. The ability to safely practice exploits, analyze network traffic, and test web applications in a controlled environment accelerates learning while maintaining ethical and legal boundaries. This lab environment will serve as the primary workspace for subsequent internship tasks involving vulnerability assessment, exploit development, and security auditing.

References

Kali Linux Kali Linux Official Documentation — www.kali.org OWASP Foundation OWASP Documentation & Juice Shop — owasp.org Wireshark Wireshark User Guide — www.wireshark.org	Oracle VirtualBox VirtualBox User Manual — www.virtualbox.org Nmap Nmap Official Guide — nmap.org Docker Docker Documentation — docs.docker.com
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